MACHINE LEARNING • HEALTHCARE

Breast Cancer Prediction & Classification System

A Random Forest approach to classifying breast masses as benign or malignant



PROJECT PRESENTATION

Project Overview

What the system does and why it matters

Objective

Build a supervised machine-learning model to classify breast-mass records as benign or malignant.

Approach

Clean and prepare diagnostic features, then train a Random Forest classifier.

Outcome

The report records 96.49% accuracy on 114 held-out test samples.

Important context

This is a decision-support research project based on diagnostic measurements—not a replacement for professional medical assessment.

Dataset at a Glance

Breast Cancer Wisconsin (Diagnostic) Dataset

569 samples

Patient records in the dataset

30 features

Numerical diagnostic measurements

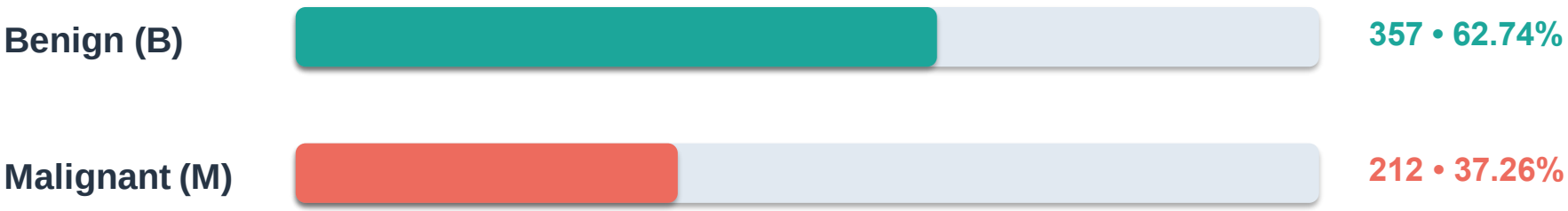
33 columns

Including ID, diagnosis and an empty export column

2 classes

Benign and malignant

Target distribution



Measurements are computed from digitized fine-needle aspirate (FNA) images of breast masses.

What the Model Learns From

Thirty numerical features describe cell-nucleus shape and texture

Size & boundary

Radius • Perimeter • Area

Surface & shape

Texture • Smoothness • Symmetry

Contour complexity

Compactness • Concavity • Concave points

Pattern detail

Fractal dimension

Each of the 10 core measurements is represented as mean, standard error, and “worst” value.

Data Preparation Pipeline

Steps described in the project documentation

01

Remove ID

Drop non-predictive identifier

02

Remove empty column

Discard 100% missing export artifact

03

Encode diagnosis

Malignant = 1 • Benign = 0

04

Split data

80% train (455) • 20% test (114)

05

Standardize

Fit scaling on train; apply to test

Reproducibility & leakage prevention

A fixed random seed supports reproducibility. Scaling parameters are learned from training data only.

MODEL ARCHITECTURE

Random Forest Classifier

An ensemble of 100 decision trees



Why Random Forest? Captures non-linear patterns and combines multiple trees to reduce reliance on any single tree.

Model Performance

Evaluation on 114 held-out test samples

TEST ACCURACY

96.49%

4 errors out of 114 samples

Class	Precision	Recall	F1	Support
Benign	0.96	0.99	0.97	71
Malignant	0.98	0.93	0.95	43
Macro avg	0.97	0.96	0.96	114
Weighted avg	0.97	0.96	0.96	114

Malignant recall (0.93) matters: some malignant cases were missed, so clinical validation is essential.

Interpreting the Results Responsibly

Accuracy is useful—but not the whole story in medical classification

Precision

Malignant precision = 0.98. Most cases predicted malignant were malignant in this test set.

Recall / sensitivity

Malignant recall = 0.93. The model identified most, but not all, malignant cases.

Next validation step

Evaluate on additional data and assess clinical risks before any real-world use.

Scope note

The reported metrics come from a single held-out split. They do not establish clinical readiness, generalization to every population, or suitability for independent diagnosis.

Future Enhancements

Priorities identified in the project documentation

1. Hyperparameter tuning

Use Grid Search or Randomized Search to optimize tree depth, leaf size and split criteria.

2. Compare algorithms

Benchmark SVM, Gradient Boosting and Deep Neural Networks against Random Forest.

3. Explain predictions

Analyze feature importance and attribution to understand which measurements influence predictions.

Longer-term direction: stronger validation, clearer interpretability, and careful assessment before clinical deployment.

CONCLUSION

A promising machine-learning baseline

The project combines data cleaning, feature standardization and a 100-tree Random Forest to classify benign and malignant cases.

01 **569 diagnostic samples**

02 **96.49% reported test accuracy**

03 **Future work: tuning, comparisons and interpretability**

Thank you • Questions?